

Matthew Millward

+44 XXXX XXXXXX | MMillward2012@gmail.com | [LinkedIn Profile](#) | [Portfolio Website](#)

EDUCATION

Durham University – BSc Mathematics

Graduating 2026

Predicted First Class Honours (85.0% average after Year 2)

- Third-year modules: Stochastic Processes, Mathematical Finance, ML & Neural Networks, Decision Theory
- Final Year Dissertation: Applying reinforcement learning to options market making and dynamic hedging strategies, using policy gradient methods and Deep-Q-learning to optimise sequential trading decisions

Sir Graham Balfour

2016 – 2023

*A-Levels: A*A*A*A*A in Maths, Chemistry, Physics, Further Maths, and an EPQ (Extended Project Qualification)*

PROFESSIONAL WORK EXPERIENCE

Google DeepMind - AI Research Intern

June – August 2025

Project: *Real-Time Financial NLP for Low-Latency Trading Systems*

- Achieved **1.5ms inference latency** (from 224ms baseline) optimising TinyBERT for **financial sentiment classification**, scaling to **2,040+ samples/sec** throughput through FP16 precision, dynamic batching, and **GPU memory optimisation** on NVIDIA T4 infrastructure
- Improved model **F1 score** from **79% to 89%** (surpassing FinBERT **baseline of 84%**) via confidence-weighted fine-tuning on **Financial PhraseBank** data, integrating **explainability frameworks** to prioritise misclassified predictions
- Designed automated benchmarking system evaluating 5 transformer architectures across CPU/GPU/TensorRT configurations, measuring **latency percentiles** and **memory efficiency** to identify optimal speed-accuracy trade-offs for production deployment
- Awarded "**Best Research**" at TAROS 2025 international AI conference

PERSONAL/ACADEMIC PROJECTS

High-Performance Options Analytics Platform for Strategy Research (Python)

- Built a Monte Carlo engine (Numba JIT + prange) for **200M+ simulations/sec**, validating Black-Scholes convergence to **99.99% accuracy** (<\$0.0007 absolute error on an \$8 option) for precise option pricing and hedging
- Identified **put-call parity violations of 0.5–2.0%**, achieving **60% execution rate** and quantifying the gap between theoretical arbitrage and real-world capture
- Simulated 1,000+ delta-hedging paths with discrete rebalancing; quantified **±\$2–5 daily P&L error**, halving risk with hourly rebalancing and capturing fat-tail skew
- Backtested 5 years of SPY (1,000+ scenarios), isolating 60% profitable months vs theoretical **Sharpe 2.5–4.0** and realised **Sharpe 1.2–2.0 after costs**

Flight Price Prediction via Temporal Deep Learning (Python)

- Developed 30-day price forecasting model achieving **29% median cost reduction** (\$70 per booking, \$240 median fare) vs immediate purchase on 12,291 domestic and international flights; delivered **61% directional accuracy** and **±2 day median error** in identifying optimal booking timing
- Corrected temporal fusion transformer's variance underestimation through a gradient-boosted residual layer with volatility features (Z-scores, price spreads, momentum); **improved MAE** on extreme prices (>90th percentile) from **\$107 to \$52** while maintaining **0.91 variance ratio**, avoiding mean-reversion collapse in neural regression

HACKATHONS

DurHack (Durham 24-hour Hackathon)

- **G-Research “Predict the future” Challenge (2025):** Developed XGBoost risk model predicting shark attack probability from 6 engineered features (e.g., shark density, coastal population), achieving **0.96 validation accuracy** on 1,200 historical incidents supplemented with synthetic data generation; developed interactive React globe visualising spatiotemporal probability surfaces with a temporal slider across future months
- **Route Optimisation & System Design Projects (2024/23):** Produced a Dijkstra-based route optimiser for navigating between multiple landmarks in Durham, and an e-commerce matchmaking prototype (“Tinder for Amazon”)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL
- **Libraries/Tools:** PyTorch, NumPy, Pandas, Hugging Face, Scikit-learn, SQLite, Git, Matplotlib, Excel

CERTIFICATIONS

- Academy for PhD Training in Statistics (APTS)

October – December 2024